

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 14-200-2
CABLE JOINTS FOR XLPE INSULATED CABLES 18/30 (36) kV (ARMOURED AND NON ARMOURED)	13-05-2025

EDMS 14-200-2

**SPECIFICATION
FOR
CABLE JOINTS FOR XLPE INSULATED CABLES
18/30 (36) kV (ARMOURED AND NON ARMOURED)**

Issue: May. 2025/ Rev- 2

- This revision contains option items that must be selected by EDC before bidding.

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1. SCOPE

This specification covers the minimum technical requirements for design, engineering, manufacture, testing, inspection, installation and performance of all kinds of straight through joints for medium voltage power cables, XLPE insulation, with the following different constructions:

- Three core and single core
- Armoured and non armoured
- Copper and aluminum conductors
- The joint for the level of 22 kV system (18/30 kV cable) should meet the requirements of the 36 kV Level.
- The Joint kits should be supplied completely with all relevant components
- Unless otherwise specified Joints should comply with all relevant standards described in table no.1

This specification does not cover/

- Accessories for special applications, such as aerial cables, submarine or ship cables or hazardous situations (explosive environments, fire-resistant cables or seismic conditions).
- Transition joints connecting cables with extruded insulation to paper-insulated cables are not included in the scope of this specification

2. APPLICABLE STANDARDS

The joints covered in this specification shall comply with the latest edition/amendment of the standards given in Table (1). Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply. Any deviations from the listed standards or the provisions of this specification should be clearly set out in the offer.

S. No.	Cross Reference	Description
1.	EDMS 15-301	COMPRESSION TYPE CONNECTORS For CABLES UP TO 24 kV

Table -1

S. No.	Standard No.	Description
1.	IEC 60502-2	Cables for rated voltages from 6 kV ($U_m = 7,2$ kV) up to 30 kV ($U_m = 36$ kV)
2.	IEC 60502-4	Test requirements on accessories for cables with rated voltages from 6 kV ($U_m = 7,2$ kV) up to 30 kV ($U_m = 36$ kV)
3.	IEC 61442	Test methods for accessories for power cables with rated voltages from 6 kV ($U_m = 7,2$ kV) up to 30 kV ($U_m = 36$ kV)
4.	CENELEC HD 629.1S3	Test requirements on accessories for use on power cables rated voltage from 3.6/6(7.2) kV up to 20.8/36(42) kV - Part 1
5.	IEC 60228	Conductors of insulated cables
6.	IEC 60060-1,2	High-voltage test techniques
7.	IEC 60230	Impulse tests on cables and their accessories

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8.	IEC 60885-3	Electrical test methods for electric cables Part 3: Test methods for partial discharge measurements on lengths of extruded power cables
9.	VDE 0278-629-1	Test requirements on accessories for use on power cables of rated voltage from 3,6/6 (7,2) Kv up to 20,8/36 (42) kV Part1: accessories for extruded insulation
10.	IEEE 404	Standards for Extruded and Laminated Dielectric shielded cable joints rated 2.5 kV to 500 kV
11.	IEC62677-1	Heat shrinkable low and medium voltage moulded shapes. General requirements
12.	IEC62677-2	Heat shrinkable low and medium voltage moulded shapes - part 2: methods of test
13.	IEC62677-3-101, 102, 103	Heat-shrinkable low and medium voltage ... Part 3: Specification for individual materials

3. DEFINITIONS

- Straight joint

Accessory makes a connection between two cables and suitable for use underground or in air.

- Branch joint

Accessory makes a connection of a branch cable to a main cable.

- Transition joint

Straight or branch joint makes a connection between cables having different types of insulation, and/or conductor material, and/or conductor cross section area, and/or cable types.

- Connector

A metallic device suitable to connect the cable conductors together, equipped with two lids to tightly seal the aluminum connector to keep the grease inside.

- Conductor Temperature:

The joint should be suitable for use on cables having maximum conductor temperature of 90°C at normal conditions, and maximum temperature of the cable conductors under short-circuit condition of 250°C

- Rated voltage: U_0/U (U_m)

The rated voltages $U_0/U(U_m)$ of the cable joints considered in this specification is 18/30 (36) kV.

- U_0 : is the rated power frequency voltage between conductor and earth or metallic screen for which the cable is designed (phase voltage).
- U : is the rated power frequency voltage between conductors for which the cable is designed (line voltage);
- U_m : is the maximum value of the "highest system voltage" for which the equipment may be used.

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The performance of the pre-moulded and heat-shrinkable components in the joint should be guaranteed for following environmental conditions, any differences in the guaranteed performance should be clearly set out in the offer.

Ambient temperature	-5°C to +45°C (50°C is an option according to --EDC requirements),
Maximum relative humidity	95 %
Altitude	Up to 1000 m above mean sea-level

5. SERVICE CONDITIONS

- The sunlight is very intense and the temperature of exposed 70°C.
- The maximum solar radiation intensity on a horizontal surface is 1000 w/m².
- Severe surface contamination from salt, dust, heavy industrial pollution, sandstorms, fine dust often carried in suspension in the atmosphere even on calm days, can be expected.
- Ground temperature can reach 35°C.
- Joints should be suitable for direct burial on the following conditions: -
 - a. Where the water table is very high, and the joints are thus exposed to polluted sea water / sewage water.
 - b. With laying depth of the cable up to 2.0 m
 - c. In dry sand conditions with soil thermal resistivity as per the cable specification

- Soil thermal resistivity (C°. Cm/watt)	120	200	300
- Derating factor	1.0	0.80	0.67

6. SYSTEM DETAILS

Frequency	50 Hz
Number of phases	Single / Three
Nominal voltage	22 kV
Highest voltage	24 kV
Maximum prospective Fault level	20 kA for Duration of 1 sec.

7. JOINT MATERIALS, RATING AND PERFORMANCE

- 7.1 The Joints kits should be supplied completely with all relevant components including the main insulation part(s), DIN or equivalent standards ferrules (connectors) , tinned copper mesh or Braid of sufficient length, cleaning solvent, solder , armouring (as applicable) , outer jacket tubes , and/or any special accessories required.
- 7.2 For three-core armoured cable, a galvanized steel or nonferrous wrap and adhesive tape or cotton cloth tape bedding should be used for armouring the cable joints, or any other applicable solution for the armoring layer of the cable joint.

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- 7.3 For single core armoured cable, a nonferrous or AL wrap and adhesive tape or cotton cloth tape bedding should be used for armouring the cable joints, or any other applicable solution for the armouring layer of the cable joint.
- 7.4. The material for bonding the cable screen , the cable armouring with the joint screen (copper braid or mesh) should be suitable for fault levels specified in clause 6
- 7.5 The cross-sectional area of the copper mesh or braid should be at least equal to the effective cross-sectional area of the cable copper screen with a minimum size as indicated in the following tables.

Copper Screen of joint for the Single-Phase Cable 18/30 kV

C.S.A (mm ²)	70	95	120	150	185	240	300	400	500	630	800	1000
Total Copper weight per joint (gm/m)	130	130	130	200	200	200	260	315	315	315	315	315
Min. length of the copper Baird per joint	90	90	90	90	90	90	90	100	100	100	100	100

Copper Screen of joint for the Three Phase Cable 18/30 kV

C.S.A (mm ²)	70	150	185	240	300	400	500
Total Copper weight for the 3 phases gm/m	390	390	390	390	480	675	675
Length of the copper Baird per phase cm	90	90	90	90	90	100	100

- 7.6. Each kit should be supplied with installation instructions in Arabic (and English).
- 7.7. Joints should in general be compatible with cables used byEDC
- 7.8. Testing of the joints should meet the requirements described in the relevant clause and either requirements of IEC ,or VDE, or according to CENELEC or IEEE.
- 7.9. The minimum guaranteed performance should be under Egypt service conditions mentioned in item (5)
- 7.10. The 36 kV joints should be suitable for use with the following type of cables :
18/30 kV, three / one core, underground power cable, Aluminum/ Copper stranded-compacted circular conductor, extruded semi-conducting layer, XLPE insulation, extruded semi-conducting layer well bonded, copper tape screening, filler, separation sheath ST2 (bedding) , double Steel/Al tape armour, black PVC over sheath, and according to the requirements of IEC 60502-2

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7.11. The materials of the joints and the installation instructions should be compatible with the XLPE cable materials 18/30 (36) kV, hence the cable insulation , and semi-conductive screening layers are extruded type.

7.12. Connectors material, outer diameter , die size as per latest revision of EDMS 15-301.

7.13. ADDITIONAL ACCESORIES FOR JOINT KIT

- 1- Cleaning solvent >100 c.c(m.l) + cleaning tissues(3 tissues per phase), or cleaning wipes as the following:
 - 12 wipes for 3 core / 6 for single core joint.
 - Wipe: fiber (20*20cm)
 - Solvent type: isopropyl or tetrachloroethylene
- 2- Binding tapes: -
 - A. (for Pre-moulded joints only): Two binding tapes length > 10 m for each role (used for binding 3 cores before applying joint armouring), Or One insulating tape >30 m for each role.
 - B. (for heat shrinkable joints only): A Textile tape used for wrapping the ends and middle of the steel wrap.
- 3- Tin wire solder (with flux): length > 1m, or any other applicable solution
- 4- Silicon grease: (6 Packets) each of weight ≥ 5 gm for three cores
- 5- Galvanized steel clamps: 2 pcs for binding steel wrap. (only applicable for Three Core cables with a Galvanized steel wrap armouring)
- 6- Stainless clamps: 2 pcs for binding steel wrap. (only for single Core cables, and only applicable when using an Aluminum wrap armouring)
- 7- Mastic tapes (pastes) of types required for sealing and filling, as applicable at least:
 - A. Over steel clamps (if exist), under the ends of jacket tubes over the cables, and between tubes at their overlap.
 - B. At connector area as void filler (for heat shrinkable joint only).
- 8- For 3 core cable 2 shrinkable tubes used for outer jacket should be marked with diameters before and after free shrink.
- 9- 3 pcs of sandpaper size 180 with length 40 cm, cloth type (non-easy cut) (for 3 cores)
- 10- 1 pcs of sandpaper size 180 with length 40 cm, cloth type (non-easy cut) (for 1 core)
- 11- 1 piece of sandpaper size 320 with length 40 cm, cloth type (non-easy cut)

8. MARKING AND IDENTIFICATION

All components of the joint should be distinctly and clearly marked with the manufacturer's name and reference numbers. The marking should be done before packing the component. Electrically conducting components should be marked 'conducting' at the plastic packs clearly and permanently unless they are very small to accommodate respective marking.

All components should be capable of being stored without deterioration within the temperature range of -5°C to +45°C (+50 °C as option). Components or materials, if subjected to a shelf life limitation, should have the final date of use prominently and permanently shown on all packages.

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The electrical relevant components of cable accessories must be marked with the following data:

1. Manufacturer name
2. Production indicator for traceability
3. Cross-section, diameter, or application range should be marked

Packaging as well for each cable accessory should be printed with the following information:

1. Manufacturer's name
2. Manufacturer's Type No. / catalog number
3. Purchase order number/tender
4. Cable size/diameter
5. Nominal Voltage (18/30kV) / Max. Voltage (36 kV)
6. Year of manufacture
7. Date of expiry
8. Manufacturer's batch No. or any code for tracing the quality assurance records maintained by the manufacturer.
9. Shipping Mark Guid.

* It should be possible to read all information directly

9. PACKING

- 9.1. Packing should be robust and able to withstand very rough handling during transportation. It should be protected against ingress of moisture.
- 9.2. The details of method of packing should be submitted with the offer.
- 9.3. Each kit should be supplied with a packing note stating the quantities and descriptions of components contained in the kits, and the documents described below:
- 9.4. The kits should include:
Joints Instructions of installation (Arabic and English) for the offered accessories which must include the steps of installation, the number of dies, detailed drawing of the connector explaining the number & sequence of crimping and the width of each crimping, (Each crimping should be hexagonal).

10. INSPECTION AND TESTING

10.1. ROUTINE TESTS

- 10.1.1. The routine tests should be done as per latest revision of applicable standards mentioned in table No. 1.
- 10.1.2. Tests should be done by the manufacturer on each finished product or component of accessories in the manufacturer facility to check that each product/ component meets the specified requirements.
- 10.1.3. Routine tests for the cable joints should be like those of the cables:
 - 10.1.3.1 AC dry withstand voltage test (69kV for 1 min. as per IEEE 404 or 63 kV for 5 min. as per IEC 60502-2)
 - 10.1.3.2 Partial Discharge test (the PD at 30 kV should less than 5 pc)

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10.1.4. Vendor should keep records for the routine tests

10.1.5. Vendor should keep records for the calibration of the instruments and testing facilities which using in the routine tests.

10.2. SAMPLE AND INSPECTION TESTS

10.2.1. Sample tests should be made by the manufacturer on samples of finished product or components taken from a finished product store, at a specified frequency, so as to verify that the finished product meets the specified requirements.

10.2.2. Inspection test should be done in the manufacturer/vendor facilities in attendance ofEDC representatives per each order before delivery..

10.2.3. Inspection tests should be conducted as per submitted/approved inspection plan during the tender stage,

10.2.4. the inspection plan should include a visual examination on each component, there should be no visual defects in the material, manufacture, workmanship and markings.

10.2.5. The Tenderer should provide free of charge all the testing facilities required at work.

10.2.6. The routine tests should be done as per applicable standards up on manufacturer`s type tests standard.

10.2.7. The inspection sample should be 1% of the order quantity and should not be more than 5 pcs of each joint size.

10.3. TYPE TESTING

10.3.1 Type Testing should be made to demonstrate satisfactory performance characteristics according to the latest standard to meet the intended application.

10.3.2 It is not necessary to repeat these tests, once successfully completed, unless changes are made in the materials, design or manufacturing process which might affect the performance characteristics

10.3.3 Cables used for testing should comply with IEC 60502-2 and should be of the same rated voltage as the accessories to be tested

10.3.4 Accessories should be assembled in the manner specified by the manufacturer's instructions.

10.3.5 Type tests should be done in certified independent and approved test laboratory

10.3.6 Type test for cable joints should be conducted as per IEC 60502-4 table 10.

10.3.7 Test method should be as per IEC 61442 or CENELEC standards.

10.3.8 The test values and requirements should be as per IEC60502-4 table 17.

10.3.9 Type testing complying with the full requirements of CENELEC HD629.1S3 or VDE 0278-629-1 standards both accepted too.

10.3.10 Type test certificates should be submitted with the offers (if not before submitted) to be reviewed by EDC), EDC should require its representative to assess the quality control and tests

10.3.11 The test program dates should be advised by the contractor in advance to permit witnessing

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by ...EDC.

- 10.3.12 EDC reserves its right to send inspector for that purpose, and ...EDC representative should sign the test certificates in case of acceptance.
- 10.3.13 Tests described above should not release contractor from his full responsibility towards supplies, guarantee, and compliance with this specification.
- 10.3.14 Range of approval for the type test should be as per IEC 60502-4 clause 8.

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Tests ^a		Requirements	Test methods of IEC 61442:2023	Test sequences (see Figure 2)		
				2.1	2.2	2.3
1	AC voltage dry	5 min at 4,5 U_0 No breakdown.	Clause 5	X	X	X
2	Partial discharge at ambient c	10 pC max. at 1,73 U_0^b .	Clause 8	X		
3	Impulse at θ_t c	10 impulses of each polarity. No breakdown.	Clause 7	X		
4	Heating cycles in air	Min. 30 cycles at θ_t^c , and 2,5 U_0 .	10.2	X		
5	Heating cycle in water (without over-sheath damage)	Min. 3 cycles at 2,5 U_0 , no breakdown. Insulation resistance conductor to screen 10^3 M Ω minimum; screen to water 50 M Ω minimum.	10.3	X		
6	Heating cycles in water	Min. 30 cycles at θ_t^c , and 2,5 U_0 , water height 1 m.	10.3	X		
7	Partial discharge c at θ_t c and ambient temperature	10 pC max. at 1,73 U_0^b .	Clause 8	X		
8	Thermal short-circuit (screen) c	Two short-circuits at I_{sc} of the cable screen. No visible deterioration.	Clause 11		X	X
9	Thermal short-circuit (conductor) c	Two short-circuits to raise conductor to θ_{sc} of the cable.(No visible deterioration.)	Clause 12		X	X ^d
10	Dynamic short-circuit e	One short-circuit at I_d . No visible deterioration.	Clause 13			X ^d
11	Impulse at ambient	10 impulses of each polarity. No breakdown.	Clause 7	X	X	X
12	AC voltage dry	5 min at 4,5 U_0 . No breakdown.	Clause 5	X	X	X
13	Partial discharge at ambient	10 pC max. at 1,73 U_0^b .	Clause 8	X		
14	Examination according to Annex C	Shall be documented in the report.	—	X	X	X

^a The tests shall be carried out at ambient temperature, unless otherwise specified.

^c For transition joints (extruded insulation to extruded insulation), the testing parameters are those for the lower rated cable.

^d The thermal short-circuit may be combined with the dynamic short-circuit.

^e Applies to three-core cables only. For the value of the current I_d , see 9.2. Due consideration shall be given to ensure that the installation arrangement is representative for the application. The value of the class shall be declared by the manufacturer, in accordance with 9.2.

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10.3.15 Extension of approval to a range wider than that given above should be achieved by carrying out the additional tests specified at IEC 60502-4 listed in Table 6 for smaller and larger conductor cross sectional areas of the required range.

Table 6 - Extension of compliance and qualification scheme for non-range-tracking joint

Accessory size tested	Test sequences	Samples	Range of cross-sections qualified
Size < 95 mm ²	Table 14	2 (1- core) 1 (3- core)	Extension down to and including the size tested
Size > 300 mm ²	2.1 ^a	2 (1- core) 1 (3- core)	Extension up to and including the size tested

^a Test sequences refer to Table 10

Table 14 – Additional tests for smaller conductor cross-sectional areas

Tests ^a		Requirements	Test methods of IEC 61442:2023	Test sequences (see Figure 1, Figure 2, Figure 3)		
				1.1 ^b	2.1 ^c	3.1 ^d
1	AC voltage dry	5 min at 4,5 U ₀ . No breakdown nor flashover.	Clause 5	X	X	X
2	Partial discharge at ambient	10 pC max. at 1,73 U ₀ ^e	Clause 8	X	X	X
3	Impulse	10 impulses of each polarity. No breakdown nor flashover.	Clause 7	X	X	X
4	Heating cycles	No breakdown nor flashover. Joints, screened and unscreened separable connectors. Min. 10 cycles ^f in water at θ_t ^g and 2,5 U ₀ .	10.2 and 10.3	X	X	X
5	AC voltage dry	5 min at 4,5 U ₀ . No breakdown nor flashover.	Clause 5	X	X	X
6	Partial discharge at θ_t ^g , and ambient temperature	10 pC max. at 1,73 U ₀ ^e .	Clause 8	X	X	X
7	Examination according to Annex C	Shall be documented in the report.	–	X	X	X

^a The tests shall be carried out at ambient temperature, unless otherwise specified.

^c Joints: test half the number of samples given in Figure 2.

^e Not required for accessories installed on 3,6/6 (7,2) kV cables having unscreened insulation. This requirement can also be achieved with partial discharge tests conducted at 2 U₀.

^g θ_t is the maximum cable conductor temperature in normal operation +5 K to 10 K.

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Table 17 – Summary of test voltages (see Clause 9)

Rated voltage $U_0/U(U_m)$ kV						
Test	Test voltage	3,6/6 (7,2)	6/10 (12)	8,7/15 (17,5)	12/20 (24)	18/30 (36)
Humidity and salt fog	$1,25 U_0$	4,5	7,5	11	15	22,5
Partial discharge a	$1,73 U_0$	6	10	15	20	30
Heating cycle voltage and AC voltage/500 h	$2,5 U_0$	9	15	22	30	45
AC voltage/1 min	$4 U_0$	14,5	24	35	48	72
AC voltage/5 min	$4,5 U_0$	16	27	39	54	81
Impulse (peak)	–	60	75	95	125	170

^a Not required for accessories installed on 3,6/6 (7,2) kV cables having unscreened insulation. This requirement can also be achieved with partial discharge tests conducted at $2 U_0$.

11. TENDER DOCUMENTS AND SUBMITALS

The tenderer should submit with his offer a duly completed and signed package of all the required documents as:

- 11.1 A reference list showing that the proposed technique has been extensively used by recognized utility companies outside the manufacturer's country for 3 years (for foreign manufacturers only).
- 11.2 Copies of recently (latest) conducted type test certificates (electrical and mechanical) or qualification reports from certified independent and approved test laboratory for the offered joints that verify all the requirements according to IEC 60502-4 or VDE 0278-629-1 or CENELEC HD 629-1 standards.
- 11.3 Data sheet for the Connectors used within the accessories should include at least the following data:
 - 11.3.1 Connector material details (alloy, conductivity, chemical composition ...)
 - 11.3.2 assembly technique,
 - 11.3.3 tooling, dies and necessary setting,
 - 11.3.4 preparation of contact surfaces, if applicable,
 - 11.3.5 type, reference number and any other identification of the connector,
 - 11.3.6 connector details of the type test approval, the standard with which they comply (i.e. IEC 61238-1) or DIN
- 11.4 The Factory Acceptance Test (FAT), which carried out at manufacturer/vendor facilities on each finished product and components of accessories to check that each product and components meets the specified requirements
- 11.5 A detailed list of inspection tests that have been carried out at manufacturer's facilities in attendance of the ...EDC representative(s).
- 11.6 The name of the laboratory where tests can be conducted.

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- 11.7 Complete technical details with dimensional drawings, showing description of material, the following data at least should be included:
- 11.7.1. name of manufacturer and manufacturing location.
 - 11.7.2. type, designation, or code.
 - 11.7.3. rated voltage,
 - 11.7.4. minimum and maximum cable insulation diameters,
 - 11.7.5. minimum and maximum cable cross-sections for this rated voltage.
 - 11.7.6. applicable cable construction including the material and shape of cable conductor.
 - 11.7.7. installation instructions (reference).
- 11.8 Catalogue and Instructions of installation (Arabic and English preferred) of the offered accessories. The catalogues must include the number of dies, detailed drawing of the connector explaining the number of crimping and the width of each crimping, each crimping should be hexagonal.
- 11.9 Details of storage, shelf life under Egypt service conditions mentioned in clause 6 (It must not be >30 months).
- 11.10 Lifetime of joint should not be less than 30 years starting from the date of operation.
- 11.11 One sample of complete kit will be submitted with the offer.
- 11.12 Details of any special tool requirement.
- 11.13 Schedule of guarantee duly completed and signed by the manufacturer and vendor/Agent, otherwise the tenderer should be rejected.
- 11.14 Kit content lists stating the quantities and descriptions of components, showing dimensions of connectors galvanized steel wrap, and tinned copper mesh or Braid.

12. INSTALLATION INSTRUCTIONS:

- 12.1 The installation instructions should be clearly assigned to the respective joint according to the application of this Technical Specification.
- 12.2 All necessary steps for installing the joint are to be described in an easily readable manner including images for each step.
- 12.3 The installation instructions must be approved by ...EDC prior to tender.
- 12.4 14.4 The installation instructions must contain the required designations and sheath-opening (cutback) dimensions.
- 12.5 The joint components, the kit content list, and the respective installation instructions must be clearly and unambiguously identifiable.
- 12.6 The installation instructions should be enclosed in English and Arabic Language (where applicable) and are printed in color.
- 12.7 Any modifications to the assembly or the material, even insignificant, which have an effect on the installation instructions, must be clearly noted on each installation instruction and packaging unit for at least half a year (e.g. with a colored sticker with exact reference to which point was changed), as agreed with ...EDC.

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12.8 The revision level, the date of creation or revision must be recognizable on the installation instructions.

12.9 The final pages of each installation instructions is a full-page completely assembled cable joint in a cut-away/sectional view showing all active components of the kits.

13. TRAINING

The Vendor should provide a sufficient practical training wherever applicable for the installation and commissioning of the joints. (if required by ...EDC).

14. GUARANTEE:

The vendor shall guarantee supplies against all defects arising out of faulty design or workman ship or defective material for a period of two year from date of delivery to ... EDC stores, inspection and technical acceptance.

The vendor shall guarantee the uniformity of the products delivered with the approved samples.

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13	Mastic pastes : - A – 1 st type: type / number/ weight. each B – 2 nd type: type / number/ weight. each	----- ----- -----
14	Any other tape : type , length -----	
15	Attach details on rated short circuit current capacity : - a) Copper mesh or BRAID b) Armour c) Bonded mesh or BRAID with armour	
16	Copy of works routine test reports and if requested byEDC also batch / type test reports on the kits being supplied should be cleared byEDC prior to shipment confirm compliance.	Yes / No
17	Confirm that visual inspection and routine tests should be carried out on each kit -----	Yes / No
18	Confirm compliance to the appropriate requirements of table 2-----	Yes / No
19	Marking and packing:- a) Marking should be to clause 4.4 ----- b) Packing should be to clause 4.5 ----- c) Each kit with components should be complete in itself and separately packed ----- d) Submit detailed method of packing	Yes / No Yes / No Yes / No Submitted / not subm.
20	Guarantee confirm compliance to clause 6.3 ----	Yes / No
21	For direct technical clarification from the manufacturer submit the details mentioned below of the persons responsible : - Name :----- Designation : ----- Address : ----- Telephone : ----- FAX : -----	

We guarantee above mentioned data for our offered material forEDC
Vendor / Agent

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Table 2 SCHEDULE GUARANTEE (PRE MOULDED, ...X mm², 18 kV)

Schedule of deviations from -----

Technical specification

This schedule be filled and signed by every tenderer. If there are no deviations or reservations.

This should be explicitly stated in this schedule.

Item Number on tender	Precise details of minor deviations included in the offer.

(Manufacturer):

Signature: -----

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Table 3 GUARANTEE FOR JOINTS, HEAT SHRINKABLE, ...X mm², 18 kV

[illegible]

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 14-200-2
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8	Any special accessories required	-----
9	Type /Quantity of cleaning solvent -----	-----
10	Length of tin wire solder with flux,----- cm.	-----
11	Weight of silicon grease ----- gm.	-----
12	Binding tapes : B - Textile tape : - number / length each ----- m.	-----
13	Mastic pastes : - A – 1st type : type / number/ weight. each B – 2nd type : type / number/ weight. each	----- -----
14	Any other tape : type , length -----	-----
15	Attach details on rated short circuit current capacity: - a) Copper braid b) Armour c) Bonded braid with armour	-----
16	Copy of works routine test reports and if requested byEDC also Batch / type test reports on the kits being supplied should be cleared byEDC prior to shipment confirm compliance ----- yes / no	-----
17	Confirm that visual inspection and routine tests should be carried out on each kit ----- yes / no	-----
18	Compliance to the appropriate requirements of Table 2 yes/no	-----
19	Marking and packing a) Marking should be to clause 4.4 -----yes / no b) Packing should be to clause 4.5 ----- yes / no c) Each kit with components should be complete and separately packed ----- yes / no d) Submit detailed method of packing yes / no	-----
20	Guarantee confirm compliance to clause 6.3 ----- yes / no	-----
21	For direct technical clarification from the manufacturer submit the details mentioned below of the persons responsible: Name :----- Designation : ----- Address : ----- Telephone : ----- Telex : -----	-----

We guarantee above mentioned data for our offered material forEDC

Vendor / Agent

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 14-200-2
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Table 4 SCHEDULE GUARANTEE (HEAT SHRINKABLE, 18 kV)

Schedule of deviations from -----

Technical specification

This schedule should be filled and signed by every tenderer. If there are no deviations or reservations.

This should be explicitly stated in this schedule.

Item Number on tender	Precise details of minor deviations included in the offer.

(Manufacturer):

Signature-----

Important:

- Each tenderer should submit with his offer the above tender " specifications for cable joints x ... mm². With each page signed by the tenderer.
- This signature means that tenderer confirms verification of his offer for these specifications. Tenders not accompanied with signed specifications should not be considered.